# CYCLES OF A K-FLAT ACYCLIC COMPLEX WITH FLAT TERMS ARE FLAT

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The proof is the author's complete argument; the comparison of cycles and the application of the flatness criterion are both inside the counted span.

*Editorial note.* Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

*Editorial note.* The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

*Editorial note.* The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

*Allowed standard prerequisites (editorial).*

- K-flatness and flatness of the terms of a complex. Same chapter, the definitions of K-flatness and the flatness criterion for modules over a ringed site. Supplies the hypotheses on K^bullet and the property proved for its cycles.
- Flatness criterion by vanishing of Tor, and the acyclicity of the tensor product with an acyclic K-flat complex. Same chapter, the Tor criterion for flatness and the statement that tensoring an acyclic complex with a K-flat complex preserves acyclicity. Supplies the computation by which the cycles are shown to be flat.

*Source reference.* Original lemma 06YQ: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3120–3120. Navigation pointer invoked at proof line 3399; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 08FG: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 3360–3360. Navigation pointer invoked at proof line 3414; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Date*: stacks-project:sites-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

**Lemma 0.1.** *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $\mathcal{K}^\bullet$ be a K-flat, acyclic complex with flat terms. Then $\mathcal{F} = \mathrm{Ker}(\mathcal{K}^n \to \mathcal{K}^{n+1})$ is a flat $\mathcal{O}$-module.*

*Proof.* Observe that

$$\ldots \to \mathcal{K}^{n-2} \to \mathcal{K}^{n-1} \to \mathcal{F} \to 0$$

is a flat resolution of our module $\mathcal{F}$. Since a bounded above complex of flat modules is K-flat (Lemma 06YQ) we may use this resolution to compute $\mathrm{Tor}_i(\mathcal{F}, \mathcal{G})$ for any $\mathcal{O}$-module $\mathcal{G}$. On the one hand $\mathcal{K}^\bullet \otimes^{\mathbf{L}}_{\mathcal{O}} \mathcal{G}$ is zero in $D(\mathcal{O})$ because $\mathcal{K}^\bullet$ is acyclic and on the other hand it is represented by $\mathcal{K}^\bullet \otimes_{\mathcal{O}} \mathcal{G}$. Hence we see that

$$\mathcal{K}^{n-3} \otimes_{\mathcal{O}} \mathcal{G} \to \mathcal{K}^{n-2} \otimes_{\mathcal{O}} \mathcal{G} \to \mathcal{K}^{n-1} \otimes_{\mathcal{O}} \mathcal{G}$$

is exact. Thus $\mathrm{Tor}_1(\mathcal{F}, \mathcal{G}) = 0$ and we conclude by Lemma 08FG. □